

Week 13 — Dice & Probability

Name: ____ Date: ____

The simulation pattern: **loop the trials** → if counts the hits → **percent = count ÷ trials × 100**.
More trials → closer to the theory.

Today's words

Word	What it means
simulation	using the computer to run an experiment thousands of times
trial	one run of the experiment — one roll, one flip
experimental probability	what actually happened: hits ÷ trials
theoretical probability	what math predicts: wanted ÷ possible outcomes
law of large numbers	bigger experiments crawl toward the theory

1 • The 36-outcome grid 🎲🎲

Each cell = die A + die B. Fill in the blanks, then answer below.

A ↓ B →	1	2	3	4	5	6
1	2	3	4	5	—	7
2	3	4	5	—	7	8
3	4	5	—	7	8	9
4	5	—	7	8	9	10
5	—	7	8	9	10	—
6	7	8	9	10	—	12

Ways to make **7**: of 36 · Ways to make **2**: of 36 · Ways to make **10**: of 36

2 • Be the computer 🤖

Follow the code exactly. (No randomness — one right answer!)

```
count = 0;
for n = 2:8
    if n > 5
        count = count + 1;
    end
end
disp(count)
```

Which values of **n** does the loop visit? What prints?

3 • Experimental or theoretical? 🧪

Write **E** or **T** next to each:

- "I rolled a die 60 times and got 13 sixes — 21.7%."
- "A die has 6 equal faces, so a six should come up 1/6 of the time."
- "The computer flipped 10,000 coins; 50.3% were heads."

🧠 Brain teaser (optional — take it home, then BRING IT BACK)

Roll two real dice **30 times** tonight and tally the sums:

sum	2	3	4	5	6	7	8	9	10	11	12
tally											

Did 7 win? Did anything weird win instead? **Bring your tally next week** — we'll combine the whole class's rolls and compare them to the computer's million.